

CELCON® M90

General purpose; good optimization of properties

Celcon® acetal copolymer grade M90 is a medium viscosity polymer providing optimum performance in general purpose injection molding and extrusion of thin walled tubing and thin gauge film. This grade provides overall excellent performance in many applications. Chemical abbreviation according to ISO 1043-1: POM Please also see Hostaform® C 9021.

Product information

Part Marking Code	POM	ISO 11469
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Rheological properties

Melt volume-flow rate	8 cm ³ /10min	ISO 1133
Temperature	190 °C	
Load	2.16 kg	
Moulding shrinkage range, parallel	2.0 %	ISO 294-4, 2577
Moulding shrinkage range, normal	1.9 %	ISO 294-4, 2577

Typical mechanical properties

Tensile modulus	2760 MPa	ISO 527-1/-2
Tensile stress at yield, 50mm/min	65 MPa	ISO 527-1/-2
Tensile strain at yield, 50mm/min	10 %	ISO 527-1/-2
Flexural modulus	2550 MPa	ISO 178
Flexural stress at 3.5%	73 MPa	ISO 178
Compressive stress at 1% strain	26 MPa	ISO 604
Tensile creep modulus, 1h	2450 MPa	ISO 899-1
Tensile creep modulus, 1000h	1350 MPa	ISO 899-1
Charpy impact strength, 23°C	188 kJ/m ²	ISO 179/1eU
Charpy impact strength, -30°C	181 kJ/m ²	ISO 179/1eU
Charpy notched impact strength, 23°C	6 kJ/m ²	ISO 179/1eA
Charpy notched impact strength, -30°C	6 kJ/m ²	ISO 179/1eA
Izod notched impact strength, 23°C	5.7 kJ/m ²	ISO 180/1A
Izod notched impact strength, -30°C	5.5 kJ/m ²	ISO 180/1A
Poisson's ratio	0.4	

Thermal properties

Melting temperature, 10°C/min	166 °C	ISO 11357-1/-3
Temperature of deflection under load, 1.8 MPa	101 °C	ISO 75-1/-2
Temperature of deflection under load, 0.45 MPa	158 °C	ISO 75-1/-2
Vicat softening temperature, 50°C/h 50N	161 °C	ISO 306
Ball pressure test	150 °C	IEC 60695-10-2
Coeff. of linear therm. expansion, parallel	120 E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), normal	120 E-6/K	ISO 11359-1/-2
Thermal conductivity of melt	0.155 W/(m K)	Internal
Effective thermal diffusivity, flow	4.85E-8 m ² /s	Internal
Specific heat capacity of melt	2210 J/(kg K)	Internal

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Flammability

Oxygen index	14.9 %	ISO 4589-1/-2
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Electrical properties

Volume resistivity	8E12 Ohm.m	IEC 62631-3-1
Surface resistivity	3E16 Ohm	IEC 62631-3-2

Physical/Other properties

Humidity absorption, 2mm	0.2 %	Sim. to ISO 62
Water absorption, 2mm	0.75 %	Sim. to ISO 62
Density	1410 kg/m ³	ISO 1183
Density of melt	1200 kg/m ³	Internal

Injection

Drying Temperature	100 - 120 °C	
Drying Time, Dehumidified Dryer	3 - 4 h	
Melt Temperature Optimum	180 °C	Internal
Max. mould temperature	80 - 120 °C	
Back pressure	4 MPa	
Injection speed	low-medium	
Ejection temperature	140 °C	Internal

Additional information

Injection molding

Standard reciprocating screw injection molding machines with a high compression screw (minimum 3:1 and preferably 4:1) and low back pressure (0.35 Mpa/50 PSI) are favored. Using a low compression screw (I.E. general purpose 2:1 compression ratio) can result in unmelted particles and poor melt homogeneity. Using a high back pressure to make up for a low compression ratio may lead to excessive shear heating and deterioration of the material.

Melt Temperature: Preferred range 182-199 C (360-390 F). Melt temperature should never exceed 230 C (450 F).

Mold Surface Temperature: Preferred range 82-93 C (180-200 F) especially with wall thickness less than 1.5 mm (0.060 in.). May require mold temperature as high as 120 C (250 F) to reproduce mold surface or to assure minimal molded in stress. Wall thickness greater than 3mm (1/8 in.) may use a cooler (65 C/150 F) mold surface temperature and wall thickness over 6mm (1/4 in.) may use a cold mold surface down to 25 C (80 F). In general, mold surface temperatures lower than 82 C (180 F) may hinder weld line formation and produce a hazy surface or a surface with flow lines, pits and other included defects that can hinder part performance.

Film extrusion Standard extruders with a length to diameter ratio of at least 20:1 are

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recommended. The screw should be a high compression ratio of at least 3:1 and preferably 4:1 to assure good melting and melt homogeneity. The design should be approximately 35% each for feed and metering sections with the remaining 30% as the transition zone.

Melt temperature: 160-220 C (320-430 F)

Other extrusion

Standard extruders with a length to diameter ratio of at least 20:1 are recommended. The screw should be a high compression ratio of at least 3:1 and preferably 4:1 to assure good melting and uniform melt homogeneity. The design should be approximately 35% each for the feed and metering sections with the remaining 30% as transition zone.

Melt temperature 180-220 C (355-430F)

Profile extrusion

Standard extruders with a length to diameter ratio of at least 20:1 are recommended. The screw should be a high compression ratio of at least 3:1 and preferably 4:1 to assure good melting and melt homogeneity. The design should be approximately 35% each for feed and metering sections with the remaining 30% as the transition zone.

Melt temperature: 180-220 C (360-430 F).

Sheet extrusion

Standard extruders with a length to diameter ratio of at least 20:1 are recommended. The screw should be a high compression ratio (at least 3:1 and preferably 4:1) to assure good melting and uniform melt homogeneity. The screw design should be approximately 35% each for the feed and metering sections with the remaining 30% as the transition zone.

Melt temperature 180-190 C (355-375 F).

Blow molding

Consult product information services.

Calendering

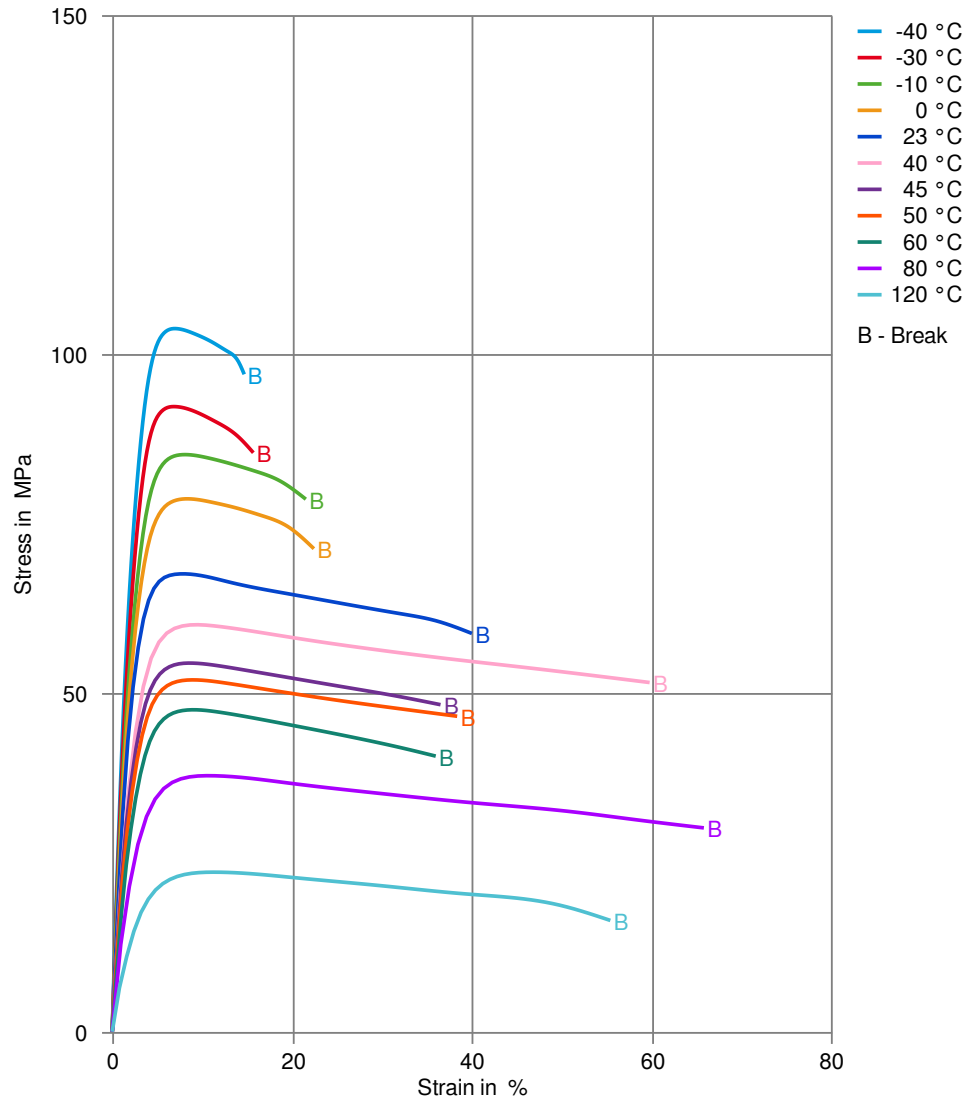
Consult product information services.

Compression molding

Consult product information services.

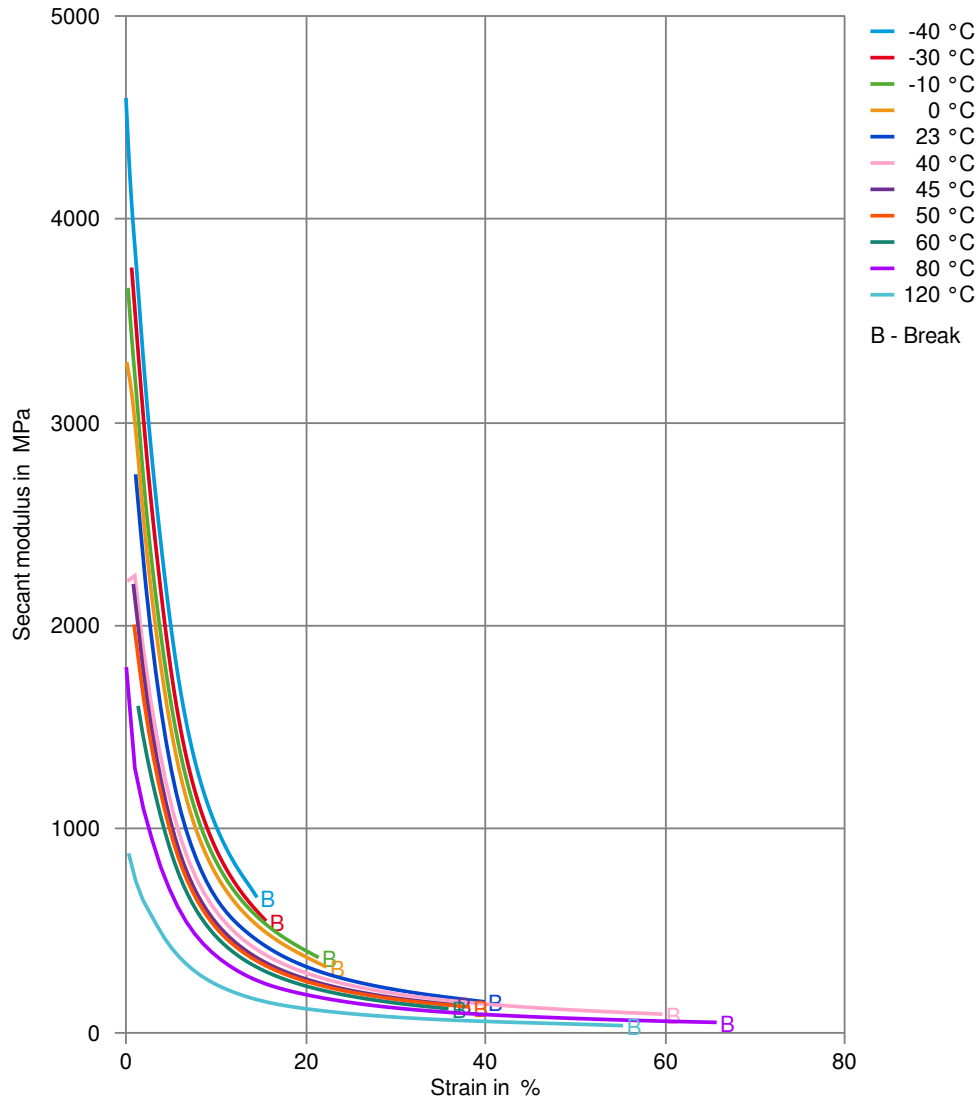
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Stress-strain



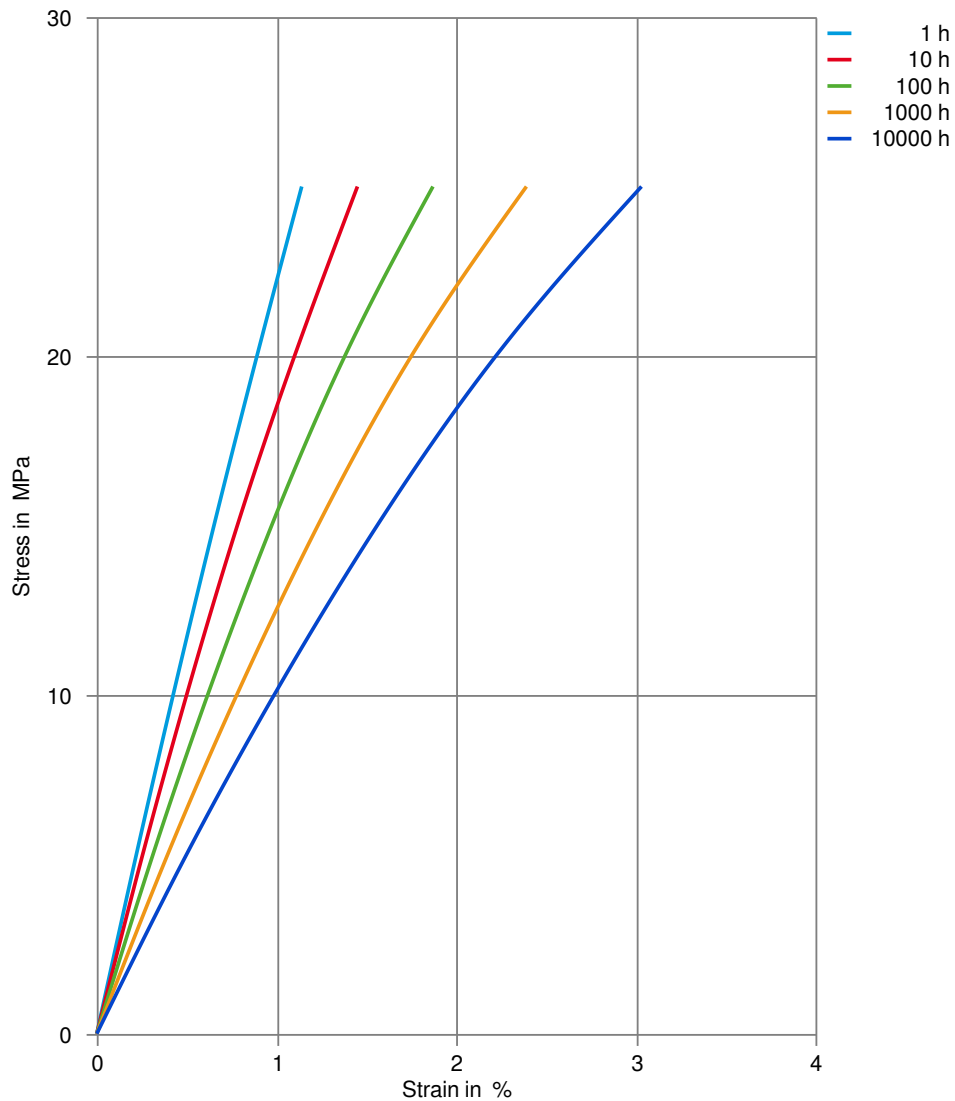
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Secant modulus-strain



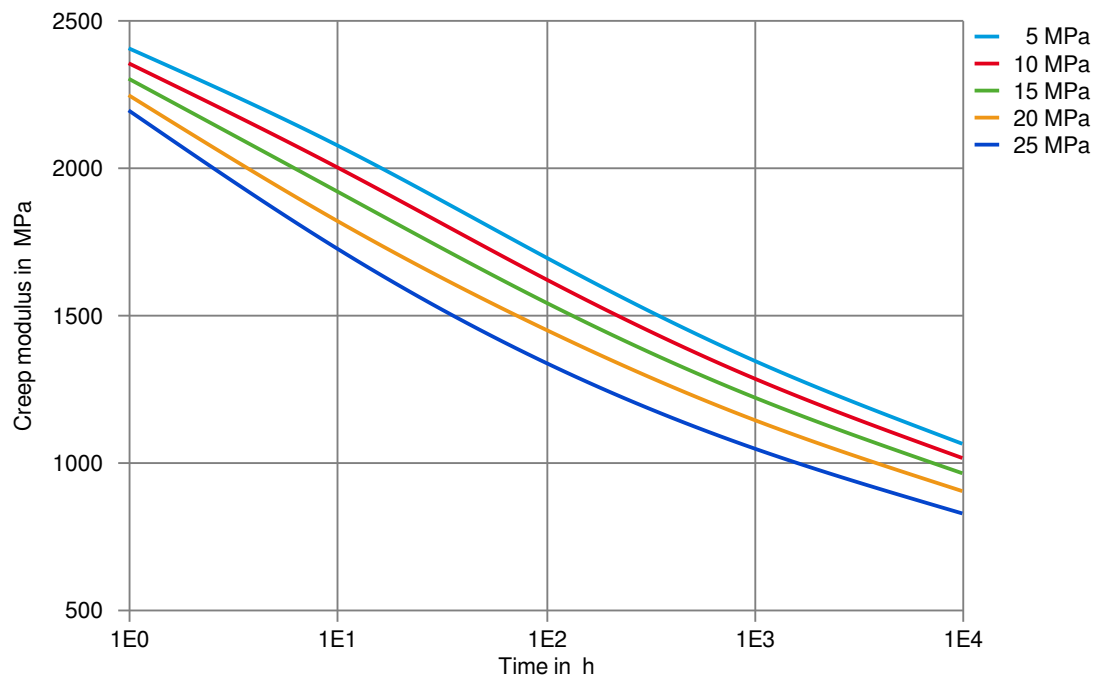
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Stress-strain (isochronous) 23°C



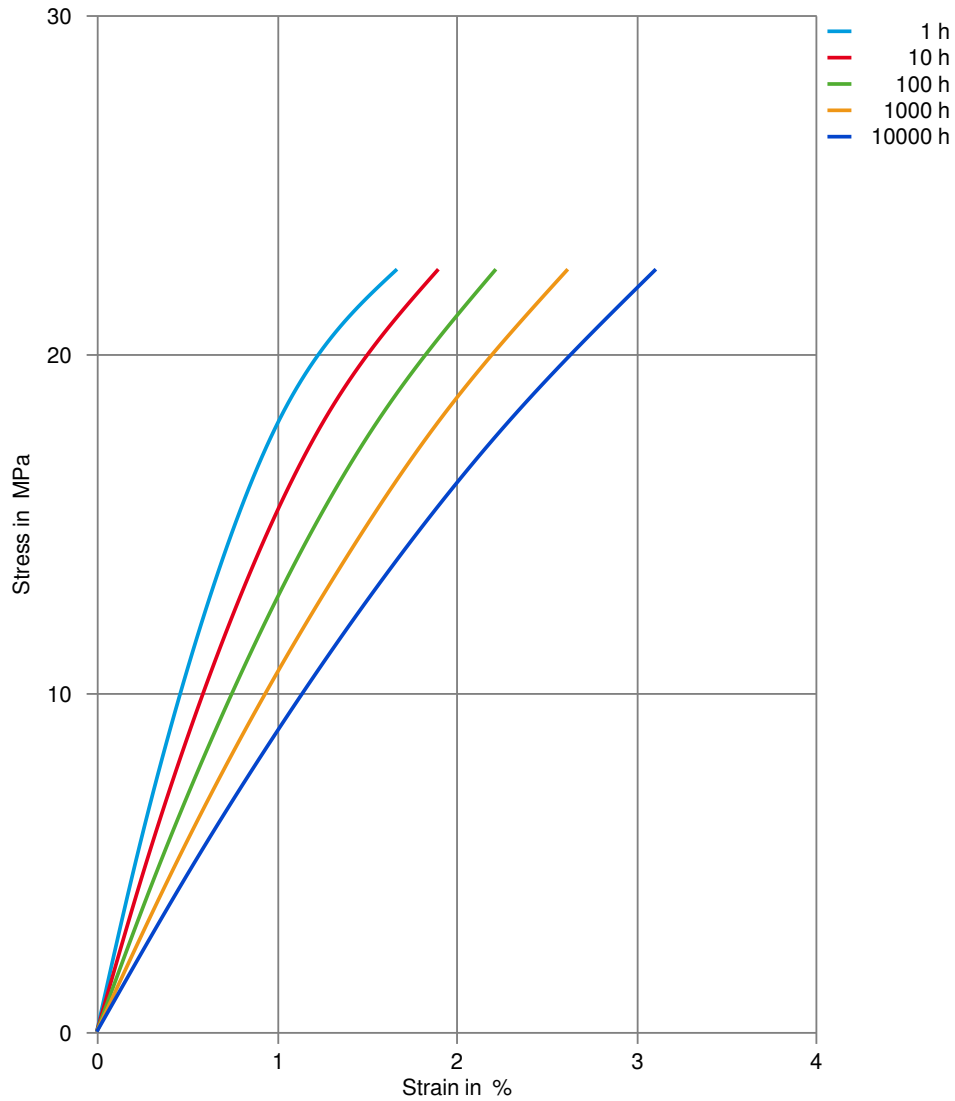
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Creep modulus-time 23°C



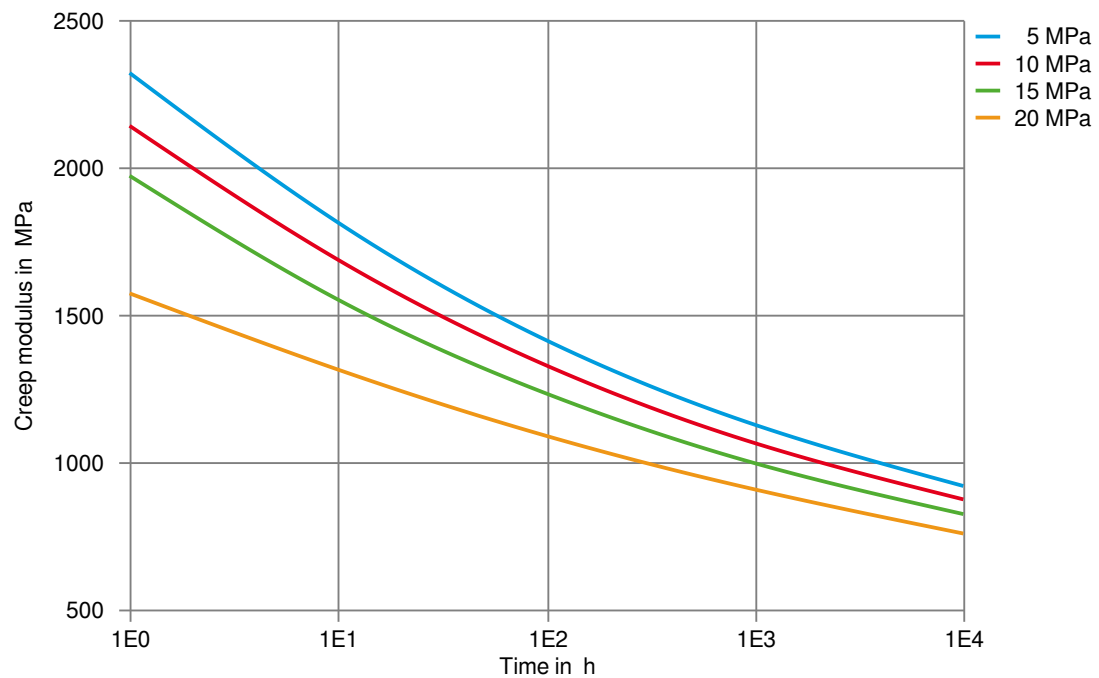
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Stress-strain (isochronous) 40°C



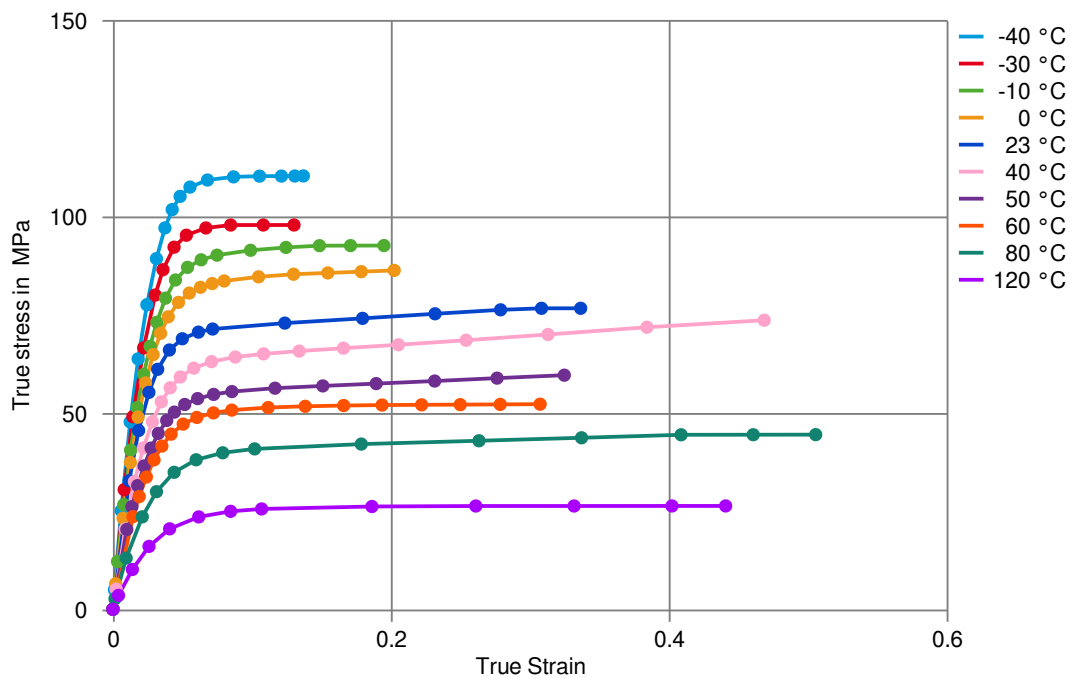
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Creep modulus-time 40°C



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True stress-strain



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Processing Texts

Pre-drying

Drying is not normally required. If material has come in contact with moisture through improper storage or handling or through regrind use, drying may be necessary to prevent splay and odor problems.

Injection molding

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Injection molding Preprocessing

Drying is generally not required because Celcon® and Hostaform® acetal copolymers are not hygroscopic nor are they degraded by moisture during processing. Excessive moisture can lead to splay (silver streaking) in molded parts. For better uniformity in molding especially when using regrind or material that has been stored in containers open to the atmosphere, recommended drying conditions are 80 C (180 F) for 3 hours. Desiccant hopper dryers are not required. Maximum water content = 0.35%

Injection molding Postprocessing

Postprocessing conditioning and moisturizing are not required. It may be necessary to fixture large or complicated parts with varying wall thickness to prevent warpage while cooling to ambient temperature.

Other Approvals

Other Approvals

OEM	Specification	Additional Information
BJEV	Q-BJEV 01.59	
Bosch	N28 BN22-O034	Natural & Black
Continental	30.5251-0367.7	

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Continental	TST N 055 54.07	
Stellantis - Chrysler	CPN 1532	Natural
Stellantis - Chrysler	CPN 1586	Black, pre-compounded or Salt & Pepper
Stellantis - Chrysler	CPN 3766	CANOD
Evergrande Auto	EGW.PL.0603-POM-CO	
Ford	WSK-M4D635-A2	Natural & Black
Geely	Q/JLY J7110235B	2018
GM	GMW22P-POM-C2	Natural & Black (replaced GMP.POM.005)
Great Wall Motor	MP05-01	
Hyundai	MS237-09, Type A	
Li Auto	Q/LiA5310020	2021 (V2)
Nissan	POM-IC2-1	
Renault	IP13g	PMR2020
Renault	UB15	PMR2020
Renault	UB03f	PMR2020
Renault	EP03a	PMR2020
Renault	EP03-3	PMR2020
SAIC Motor	SMTc 5 310 020	
Tesla	TM-1005-40	Black, Bishop USA
Tesla	TM-1005-50	Black, Bishop USA
Toyota	TSM5515G-1B	
Mercedes-Benz Group (Daimler)	DBL 5405-06-POM-C	CF2001 NATURAL

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